Exp-8: Fine-Tuned LLM Deployment for Customer Support

AIM:

To fine-tune a language model for customer support and deploy it using **Streamlit** and evaluate its performance using BLEU and ROUGE metrics.

DATASET:

Customer Service / Bank FAQ Dataset

PROCEDURE:

Step 1: Import Libraries

```
from transformers import T5Tokenizer, T5ForConditionalGeneration, Trainer, TrainingArguments
from datasets import Dataset
import evaluate
```

Step 2: Load Dataset

```
data = {
    "question": [
        "How to reset password?",
        "How to check balance?",
        "How to contact support?"
    ],
    "answer": [
        "Click on forgot password and follow instructions.",
        "Login to your account dashboard to view balance.",
        "Contact support via email or phone."
    ]
}

dataset = Dataset.from_dict(data)
```

Step 3: Preprocessing

```
def preprocess(example):
    return {
        "input_text": "Question: " + example["question"],
        "target_text": example["answer"]
    }

dataset = dataset.map(preprocess)
```

Step 4: Load Model

```
tokenizer = T5Tokenizer.from_pretrained("t5-small")
model = T5ForConditionalGeneration.from_pretrained("t5-small")
```

Step 5: Tokenization

```
def tokenize(example):
    inputs = tokenizer(example["input_text"], padding="max_length",
truncation=True)
    outputs = tokenizer(example["target_text"], padding="max_length",
truncation=True)
    inputs["labels"] = outputs["input_ids"]
    return inputs

tokenized_dataset = dataset.map(tokenize)
```

Step 6: Fine-Tuning

```
training_args = TrainingArguments(
    output_dir="./results",
    num_train_epochs=5,
    per_device_train_batch_size=2
)

trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=tokenized_dataset
)

trainer.train()
```

Step 7: Save Model

```
model.save_pretrained("model")
tokenizer.save_pretrained("model")
```

Step 8: Deployment (Streamlit)

```
import streamlit as st
from transformers import T5Tokenizer, T5ForConditionalGeneration

tokenizer = T5Tokenizer.from_pretrained("model")
model = T5ForConditionalGeneration.from_pretrained("model")

st.title("Customer Support Chatbot")

query = st.text_input("Enter your question:")

if query:
    input_text = "Question: " + query
    input_ids = tokenizer.encode(input_text, return_tensors="pt")
    output = model.generate(input_ids)
    answer = tokenizer.decode(output[0], skip_special_tokens=True)
    st.write("Answer:", answer)
```

Step 9: Evaluation

```
bleu = evaluate.load("bleu")
rouge = evaluate.load("rouge")

predictions = ["Click on forgot password and follow instructions."]
references = [["Click on forgot password and follow instructions."]]

print("BLEU:", bleu.compute(predictions=predictions, references=references))
print("ROUGE:", rouge.compute(predictions=predictions, references=["Click on
forgot password and follow instructions."]))
```

OUTPUT:

The chatbot successfully responds to customer queries through a deployed web interface.

RESULT:

The fine-tuned model generates accurate customer support responses and achieves good performance based on BLEU and ROUGE metrics.